Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the f1 score during training.

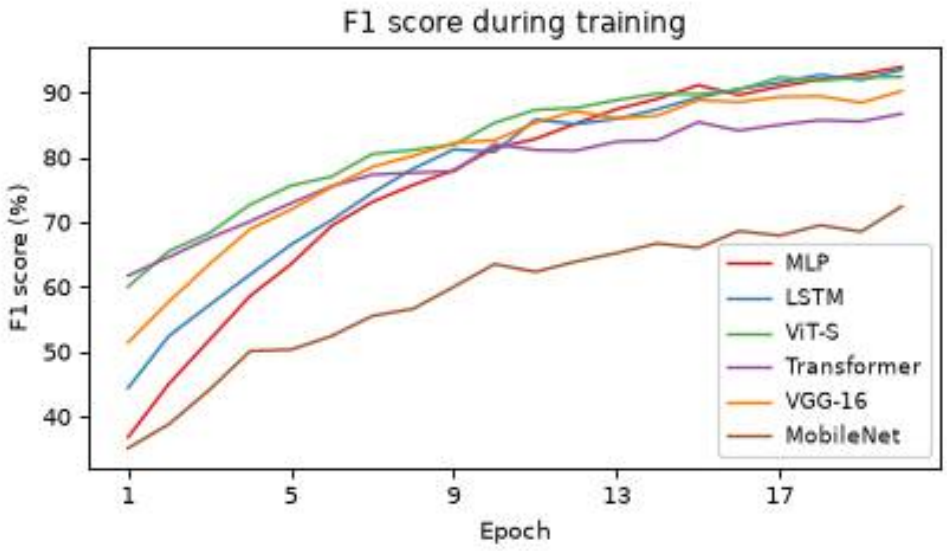


Figure 1: F1 score during training for the models in this experiment.

At epoch 3, Transformer achieves 64.3% F1 score. At epoch 8, ViT-S achieves 77.6% F1 score.

Figure 2 shows the accuracy during training.

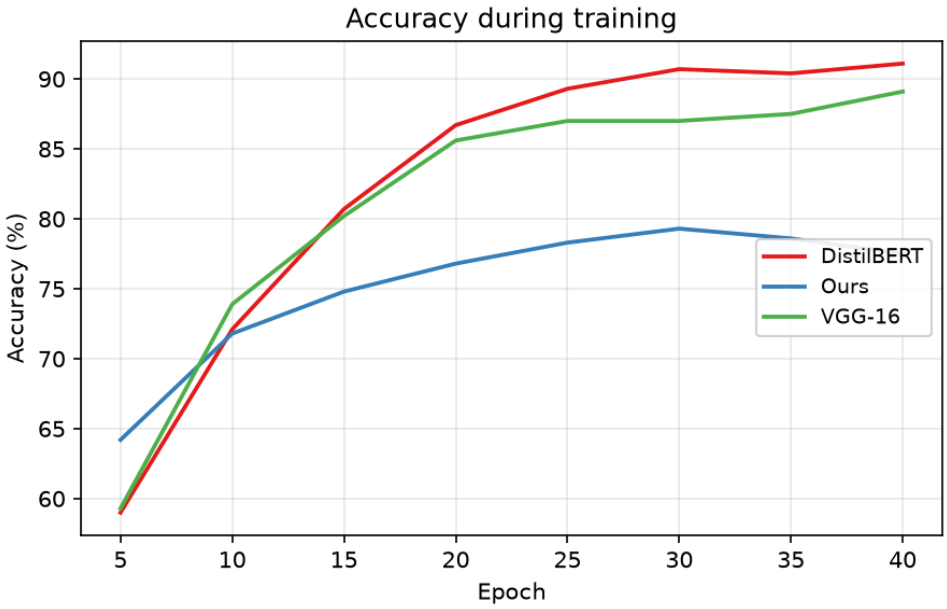


Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, from epoch 35 to epoch 40, the accuracy of DistilBERT improves by a relative 1%. As shown in Figure 2, from epoch 30 to epoch 35, the accuracy of VGG-16 improves by a relative about 1%.

Figure 3 shows the accuracy during training.

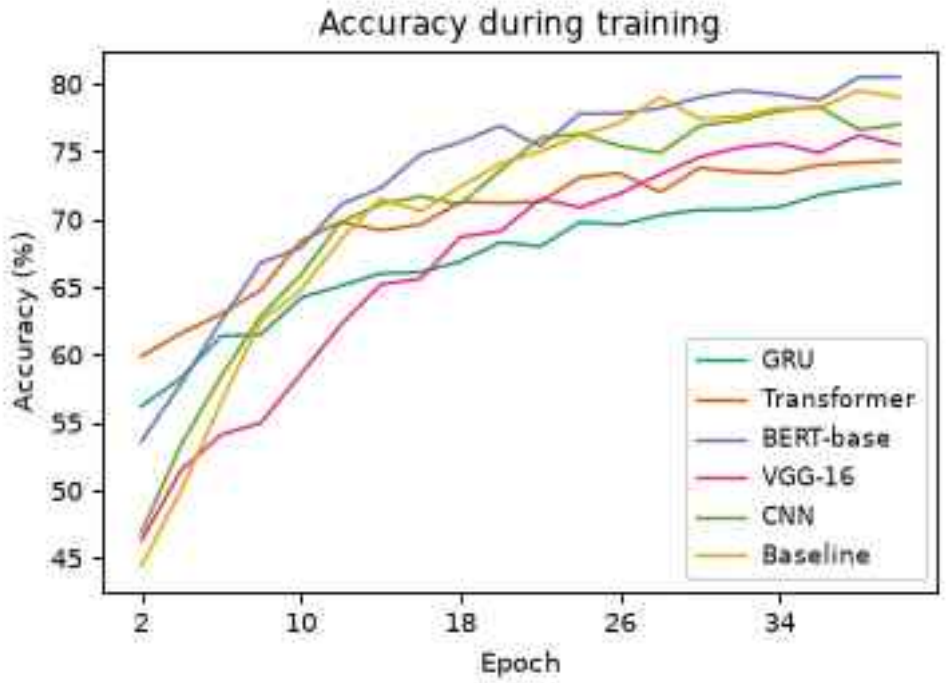


Figure 3: Accuracy during training for the models in this experiment.

At epoch 40, Baseline is about 4 points better than VGG-16. As shown in Figure 3, from epoch 8 to epoch 20, the accuracy of Transformer improves by a relative 9.9%.

Figure 4 shows the validation accuracy during training.

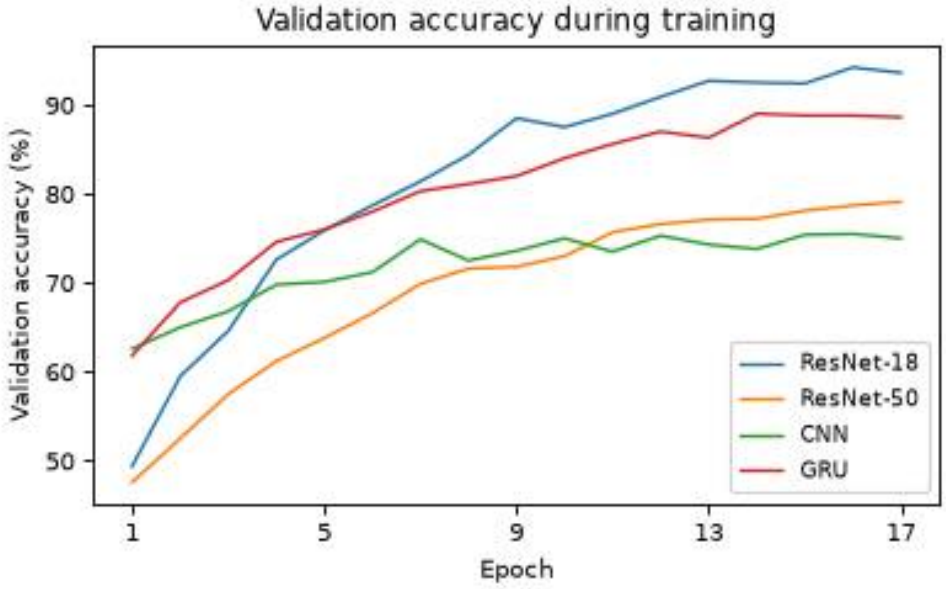


Figure 4: Validation accuracy during training for the models in this experiment.

As shown in Figure 4, CNN outperforms ResNet-18 by 2.2 percentage points at epoch 3. As shown in Figure 4, GRU reaches 70.0% validation accuracy at epoch 7. As shown in Figure 4, from epoch 9 to epoch 10, the validation accuracy of GRU improves by a relative about 2%. As shown in Figure 4, at epoch 17, CNN achieves 100.6% validation accuracy.

3 Discussion

SVM reaches 82.8% accuracy on STL-10. We note that the training was stable for all models.